Computer Science & IT



Fundamental of C Language

Control Flow Statement

Lecture No. 02



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Recap of Previous Lecture



- Topic — Control flow statement
- Topic — if, if else, if else if
- Topic — Nested if
- Topic — Ternary operator
- Topic

Slide

Topics to be Covered



- Topic — Switch
- Topic — Iterative statement
- Topic — for loop
- Topic
- Topic

Slide



Conditional Statement

Conditional Statements in C

- If-else
  - If
  - If-else
  - If-else if
  - Nested if
- (condition) ? true : false
- Switch

If

```
if(condition)
{
      //true
}
```

If-else

```
if(condition)
{
      //true
}
else
{
      //false
}
```

If-else if

```
if(condition 1 )
{
      //true
}
else if(condition 2)
{
      //true
}
else
{
}
```

Nested if

```
if(condition 1)
{
   if(condition)
      {
      }
   else
      {
      }
}
else
{
   if(condition)
      {
      }
   else
      {
      }
}
```

Switch

```
switch expression
{
      case 1;
      break;
      case 2;
      break;
      case 3;
      break;
      default;
}
```

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Switch statements

* Switch statement allows an expression to be tested for equality (=) against list of value.

* Each value in list is called a case



Switch statements

```
Switch (expression){
    case constant expression :
        statements; break;
    case constant expression :
        statements; break;
    .
    .
    .
    default: statements, break;
}
```

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Switch Syntax

```
switch(expression){
    case constant-expression  :
       statement(s);
       break; //optional
    case constant-expression  :
       statement(s);
       break; //optional

    // you can have any number of case statements.
    default : //Optional
       statement(s);
}
```

break Statement couses control transfer to outside of enclosed block of Switch Statement



Switch Syntax

```
switch(expression){
    case constant-expression  :
       statement(s);
       break; //optional
    case constant-expression  :
       statement(s);
       break; //optional

    // you can have any number of case statements.
    default : //Optional
       statement(s);
}
```

Default If None of the case matches then default is executed



Switch statements

```
switch(2+3-1) {                    5-1 = 4

    case 4   printf("case 4"); break; // case IV is true
                                       Case 4
    case 5:  printf("case 5"); break.

    case 6 : printf("case 6"), break.

    default: printf("Default case"), break.
}
```

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Switch statements

```
switch(2+3) {
    case 4  printf("case 4"); break;
    case 5: printf("case 5"); break. // case 5
    case 6: printf("case 6"); break. // print case 5
    default: printf("Default case"); break.
}
```

2+3 = 5

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Switch statements

```
switch ( 6)     {
        case 4    printf ("case 4"); break;
        case 5:   printf ("case 5"); break.

        case 6 : printf ("case 6"), break.  // case 6 is true
                                            // print  6
        default :  printf (" Default case"),  break.
}
```

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Switch statements

```
int a=4;
switch ( a+2 ) {
    case 4   printf("case 4"); break;
    case 5:  printf("case 5"); break.
    case 6 : printf("case 6"), break.   // case 6 is true
    default: printf("Default case"), break.   // print case 6
}
```

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If No break statement

if No break statement is present then all subsequent cases evaluates true until break encounter or end of block.



Question

```
#include <stdio.h>
      int main(){
       int a = 5;
        switch(a){
                  case 4: printf("%d",5);
                  case 5: printf("%d",5);
                  case 6: printf("%d",6);
        }
      }
```

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Switch statements

```
int a=4;
switch( a   . ) {
    case 4   printf("case 4");
    case 5:  printf("case 5");
    case 6 : printf("case 6");
    default: printf("Default case");
}
```

print - case 4 case 5 case 6
Default case

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Switch statements

```
int a=4;
switch( a   . ) {
    case 4   printf("case 4");
    case 5:  printf("case 5");
    case 6 : printf("case 6"); break;
    default : printf("Default case");
}
```

\n

Case4 case5 case6

← default will Not execute



What is Iterative Statement ?

print Name 100 times

```
printf ("Name");
printf ("Name");
    :
    :
printf ("Name");
```



Second Way is loop

Print My name 100 times

for loop

while loop

do while loop

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For loop: Syntax

Print My name 100 times

```
#include<stdio.h>
int main(){

    int i
    for( i=1; i<=100, i++) {
        printf("My Name").
    }
    return 0;
}
```

Initialization

Condition

Iteration

i++ = i = i+1

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For loop: Syntax

~~All Three Parts are mandatory~~

Initialization : when for loop starts variable initialized only once.

Every time before entering condition is checked

Condition checking : if condition is true then the body of loop will execute.

Body of loop will be executed

Increment : Increment variable i for Next iteration

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For loop: Syntax

Print My name 100 times

```
#include<stdio.h>
int main(){

    int i
    for( i=1; i<=5 , i++) {
        printf("My Name").
    }
    return 0;
}
```

i	i<=5	Myname	i++
i=1	1<=5	Myname	2
i=2	2<=5	My Name	3
i=3	3<=5	My Name	4
i=4	4<=5	My Name	5
i=5	5<=5	My Name	6

6<=5 ← false

exit from loop

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Decremental for Loop

```
for ( i=100, i>=1; i--)
    printf ("My Name").
```

i=100 100>=1 MyName 99

i=99 99>=1 MyName 98

:

i=1 1>=1; MyName 0

0>=1 exit

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For Loop

```
#include<stdio.h>
int main() {
    int i;
    for (i=10; i<=9; i++)
        printf("My Name");
    return 0;
}
```

i=10, 10<=9

↑

false

loop terminate



Question

```c
#include <stdio.h>
int main(){
    int i=-1;
    for(++i;i++;i++)
        printf("Name");
    return 0;
}
```

Number of times loop is Executed

(A) 0

(B) 1

(C) Infinite loop

(D) 2

i++

Exit condition Never gets true then

Infinite loop



2 mins Summary

Topic — Switch

Topic — Iterative statement - for loop

Topic

Topic

Topic

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THANK - YOU

